AIORI HACKATHON 3

THEME: Cybersecurity

INCIDENT TIMELINE AND EVIDENCE COLLECTION FOR SMALL TEAMS

Turning scattered logs into a clear incident story.



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PROBLEM DESCRIPTION

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The Challenge of Investigating Security Incidents

Security evidence is scattered across different logs, formats, and timestamps. For small teams, reconstructing an incident manually is slow and difficult.

```
/var/log/auth.log – Sample System Logs

2026-10-13T09:14:02Z  INFO   sshd[1234]: Connection opened from 192.168.1.45 port 52432
2026-10-13T09:14:08Z  WARN   sshd[1234]: Failed password for admin from 192.168.1.45 port 52432
2026-10-13T09:14:11Z  WARN   sshd[1234]: Failed password for admin from 192.168.1.45 port 52432
2026-10-13T09:14:16Z  WARN   sshd[1234]: Failed password for admin from 192.168.1.45 port 52432
2026-10-13T09:14:29Z  INFO   sshd[1234]: Accepted password for admin from 192.168.1.45 port 52432
2026-10-13T09:15:03Z  ALERT  sudo[2345]: admin : TTY=pts/0 ; PWD=/home/admin ; USER=root ; \
                                  COMMAND=/usr/bin/apt update
2026-10-13T09:15:27Z  INFO   kernel: audit: type=USER_CMD msg=audit(1697181327.123:420): \
                                  exe="/usr/bin/apt" argc=2
2026-10-13T09:16:10Z  INFO   CRON[5678]: (root) CMD (/usr/local/bin/backup.sh)
2026-10-13T09:18:44Z  ERROR  app[3456]: Failed to connect to database: timeout
2026-10-13T09:20:01Z  INFO   nginx[6789]: 192.168.1.45 - - [13/Oct/2026:09:20:01 +0000] \
                                  "GET /admin HTTP/1.1" 200 5321
```

Illustrative system logs (synthetic)

Four key problems:

1. Scattered Log Data
Security events are recorded in different files and systems, making it difficult to connect related activities.

2. Inconsistent Timestamps
Different time zones and incorrect system clocks make it difficult to arrange events in the correct order.

3. Time-Consuming Manual Analysis
Investigators must manually search, compare, and organize log entries to reconstruct an incident.

4. Evidence Integrity & Missing Records
Original evidence must remain unchanged, while unreadable or unparsed records must be identified rather than ignored.

The core problem: Small teams need a reliable way to turn scattered logs into a clear, trustworthy incident timeline.

BACKGROUND

Understanding Digital Footprints

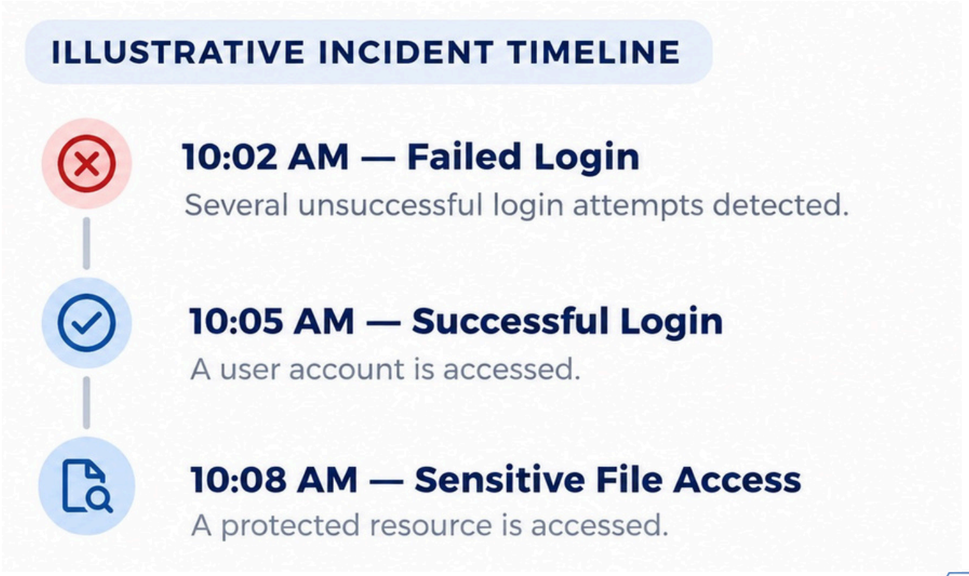
Every action on a computer system leaves behind a digital record called a log. These records help investigators understand suspicious activities, identify possible threats, and reconstruct the sequence of events during a security incident.

Main content:

System Logs
Record activities such as user logins, failed authentication attempts, and system events.

Application Logs
Capture application activity, including web requests, errors, and access to resources.

Incident Timeline
Combines records from different sources into a chronological sequence to reveal what happened and when.



Example events are fictional and used only to explain the concept.

LITERATURE & RFC REVIEW

Existing Practices That Guide Our Solution

Our approach is based on established digital forensic practices for collecting, preserving, and analyzing security evidence. These practices help ensure that incident investigations are accurate, consistent, and reliable.

Main content:

01 — NIST SP 800-86
Provides guidance on integrating forensic techniques into incident response, including evidence collection, examination, analysis, and reporting.

02 — Evidence Integrity (SHA-256)
Cryptographic hashes help verify that collected log files have not been modified during an investigation.

03 — Timestamp Standardization (UTC)
Converting timestamps to a common time reference helps combine events from systems using different time zones.

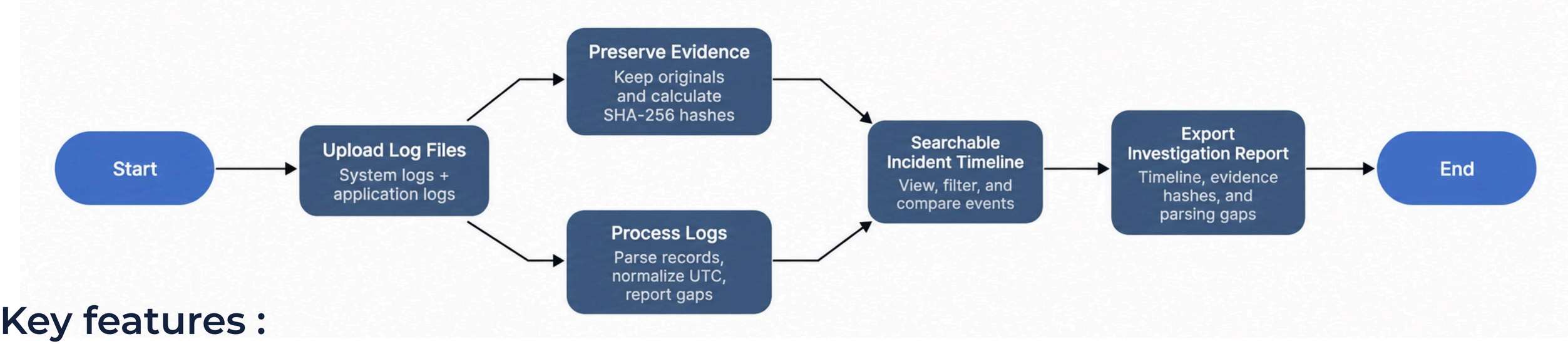
04 — Log Parsing & Analysis
Extracting important details from different log formats allows investigators to search and compare security events.

Our focus: Applying these practices in one lightweight, easy-to-use tool designed for small security teams.

PROPOSED SOLUTION

From Scattered Logs to One Clear Timeline

We propose a lightweight investigation tool that collects logs from different sources, preserves the original evidence, and organizes security events into a single searchable timeline.



Key features :

- Multi-format support: Handles at least two log formats.
- Evidence protection: Preserves original files and records hashes.
- Time correction: Converts timestamps to UTC and accounts for clock differences.
- Transparent analysis: Identifies records that could not be parsed.
- Incident reporting: Exports findings in a readable report.

SOLUTION OPTIMIZATION

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Manual Investigation vs. Our Proposed Solution

Our solution aims to reduce manual effort, improve evidence reliability, and simplify incident reconstruction through automated log processing and timeline generation.

Comparison table

Parameter	Manual Investigation	Our Proposed Solution
Log Collection	Logs reviewed separately	**Multiple log formats handled together**
Event Ordering	Events arranged manually	**Automatic UTC-based timeline**
Evidence Integrity	Requires separate verification	**Original files preserved with SHA-256 hashes**
Parsing Issues	Unreadable entries may be overlooked	**Parsing gaps identified and reported**
Searching Events	Manual searching across files	**Searchable, filtered event timeline**
Incident Reporting	Reports prepared manually	**Structured report export**

SOLUTION ARCHITECTURE & DESIGN

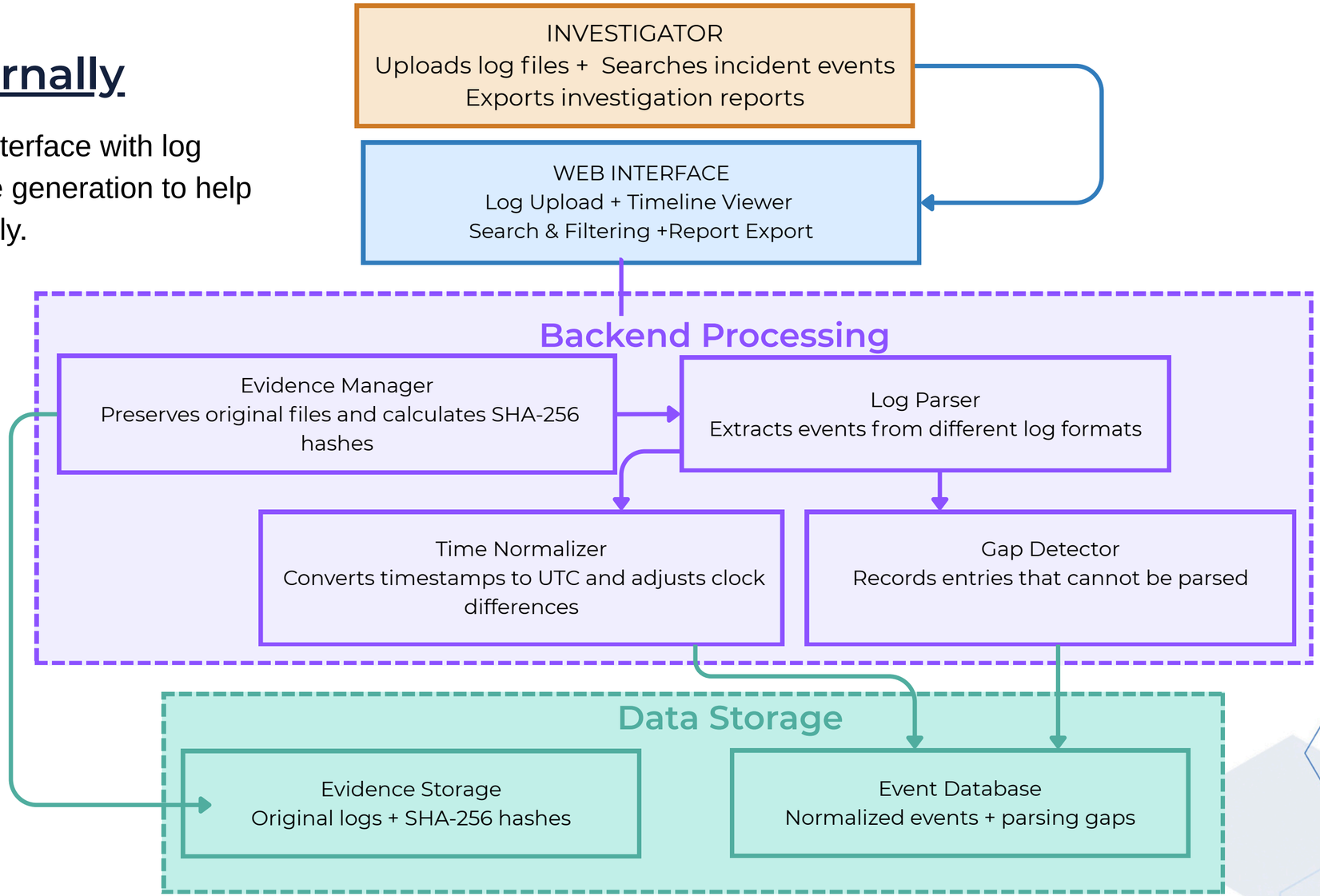
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How Our System Works Internally

Our proposed system connects a simple web interface with log processing, evidence preservation, and timeline generation to help investigators analyze security incidents efficiently.

- **Web Interface:** Upload logs and view results.
- **Evidence Manager:** Preserve originals and generate SHA-256 hashes.
- **Log Parser:** Extract events from different logs.
- **Time Normalizer:** Convert timestamps to UTC and correct clock differences.
- **Gap Detector:** Identify unparsed records.
- **Storage Layer:** Store original and processed data separately.



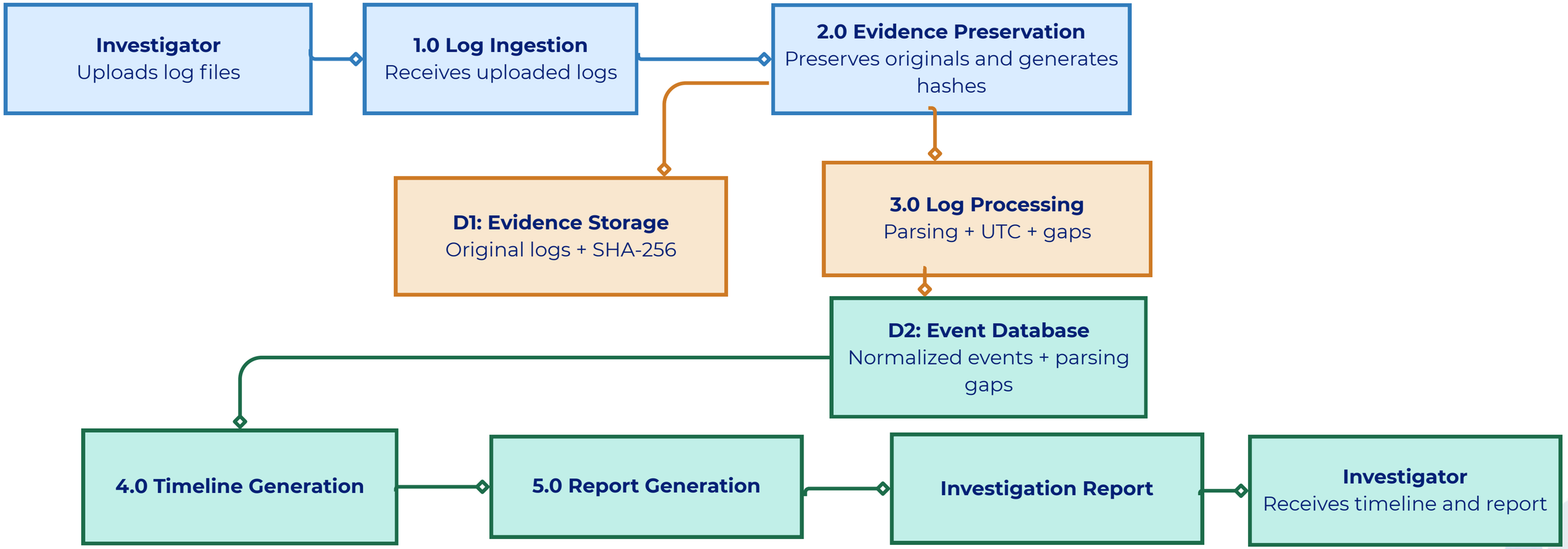
DATA FLOW DIAGRAM

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From Raw Logs to Investigation Results

The system transforms raw log files into structured, time-ordered events while preserving original evidence and recording parsing errors.



Raw logs are preserved as evidence, processed into normalized events, and transformed into a searchable timeline and investigation report.

DEPLOYMENT TIMELINE

Our Implementation Roadmap

Our development plan follows four phases, from building the core log-processing system to testing and final deployment.

Phase 1 — Foundation
Define requirements, select log formats, and design the system architecture.

Phase 2 — Core Development
Implement log ingestion, evidence preservation, SHA-256 hashing, and parsing.

Phase 3 — Timeline & Reporting
Build UTC normalization, gap detection, searchable timelines, and report export.

Phase 4 — Testing & Deployment
Test three synthetic incidents, compare triage time against manual investigation, and prepare deployment.

BIBLIOGRAPHY

Our Implementation Roadmap

Our development plan follows four phases, from building the core log-processing system to testing and final deployment.

01 — NIST SP 800-86
Guide to Integrating Forensic Techniques into Incident Response.
National Institute of Standards and Technology, 2006.
Official publication

02 — NIST FIPS 180-4
Secure Hash Standard (SHS).
National Institute of Standards and Technology, 2015. Reference for SHA-256.
Official publication

03 — RFC 3339
Date and Time on the Internet: Timestamps.
G. Klyne and C. Newman, 2002. Reference for timestamp formatting.
Official RFC

04 — RFC 5424
The Syslog Protocol.
R. Gerhards, 2009. Reference for structured system logging.
Official RFC

THANK YOU

Turning scattered logs into a clear incident story.

THE MISSING LINKS

Team ID: A3-PS013-TC084

- Questions & Discussion